

Aswathy M

Geospatial Analytics Student

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Thiruvananthapuram, Kerala

Profile

I'm a committed professional with a strong foundation in geological education and a concentration on data analytics, especially geoinformatics. My area of expertise is the integration of data-driven insights with earth sciences to enable well-informed decision-making and unearth important geographical information. I'm excellent at applying interdisciplinary approaches to address urgent problems in a variety of domains, from maximizing spatial data analysis to comprehending geological processes. My dedication to using data analytics and geoinformatics allows me to develop creative solutions in our dynamic world. I have a strong desire to use my abilities to make significant contributions and produce worthwhile outcomes.

Project Experience

- Project on '**Comparative Analysis of Image Encryption in Hyperchaotic Maps**'

March 2024 -

Project, Digital University Kerala Thiruvananthapuram, Kerala

Proposed a new image encryption algorithm based on 2D and 3D chaotic maps for better security level. The algorithm proposed involves generating hyper-chaotic signals from chaotic maps, the encryption through the use of XOR on bits, and the non-linear transformations. Assessed the method's efficiency based on the application of such performance indicators as histogram distribution, Shannon entropy, the Pixel dependency graph and the comparison of PSNR, SNR and SSIM. As evidenced by the results of the work, the proposed 3D system was highly secure and effective in preserving quality, with superior complexity compared to the previous version.

- Project on '**Finding a Sustainable Solution for Waterbody Management- A Case Study on Pothencode Pond**'

February 2024-

Project, Digital University Kerala Thiruvananthapuram, Kerala

Successfully executed both conventional and digital technologies to solve major ecological challenges while carrying out a sustainable waterbody management project for Pothencode Pond. Showed excellent abilities in data collecting, environmental analysis, project administration, and community involvement. Obtained notable gains in the health of the local ecology and the quality of the water.

- Project on '**Data Analytics And Energy Management**'

November 2023-December 2023

Project, Digital University Kerala Thiruvananthapuram, Kerala

I took part in a cooperative data analytics and energy management initiative. Using data analysis, energy consumption was optimized, production increased, and costs decreased. Patterns were found, energy needs were estimated, and decision-making improved. Evaluated advantages like lower costs and more productivity while considering data protection. Exhibited an aptitude for teamwork, data analysis, and sustainable energy solutions.

- Project on '**Geological Study in and Around Keezhayikkonm, Thiruvananthapuram, Kerala**'

January 2022-February 2022

Project, Sree Narayana College, Chempazhanthi Thiruvananthapuram, Kerala

Geological research was carried out in Keezhayikkonam, Aaramam Quarry, Thiruvananthapuram. Examined folds, faults, joints, unconformities, chondrite, and charnockite formations. We used toposheets, maps, and field observations to evaluate the lithology, minerals, and weathering. Improved quarry operations by using multidisciplinary methods. Worked in tandem with instructors, providing a thorough geological report that benefited geoinformatics and earth sciences.

Education

- **MSc.Data Analytics and Geoinformatics** **Thiruvananthapuram,Kerala 2023-**
Kerala University of Digital Science and Innovations Ist sem CGPA: 8.0
- **BSc Geology** *University of Kerala* **Thiruvananthapuram,Kerala 2019-2022**
Sree Narayana College, Chempazhanthu CGPA:8.57
- **Higher Secondary Education** **Thiruvananthapuram,Kerala 2017-2019**
Kerala State Higher Secondary Board CGPA:9.4
- **SSLC** **Thiruvananthapuram,Kerala 2017-2019**
Kerala Board of Public Examinations CGPA:9.8

Areas of Expertise

Data Analysis - Data Analytics- GIS - GIS Software- Remote Sensing -Power BI - Visualizing with Advanced Charts - Google Cloud Platform - Amazon Web Service- MS Office Suit - LaTeX- Python - SQL- R Programming

Online Courses & Certifications

- Google Cloud Career Launchpad Data Analyst Track -[Google](#)
- Introduction to Responsible AI -[Google](#)
- Introduction to Generative AI-[Google](#)
- Diploma in Computer Application-[Keltron Kerala](#)
- Web Development- [IIT Bombay,ACMEGRADE](#)

Volunteering

- Core Committee Member at Social Engagement Center- Digital University Kerala, Thiruvananthapuram, Kerala
- National Service Scheme Volunteer at Digital University Kerala, Thiruvananthapuram, Kerala
- Joint Convener, Reading Club - Digital University Kerala, Thiruvananthapuram, Kerala
- Science Volunteer- Global Science Festival Kerala(GSFK)
- National Service Scheme Volunteer- Sree Narayana College, Chempazhanthu,Thiruvananthapuram, Kerala
- Member of Placement Cell-Digital University Kerala, Thiruvananthapuram, Kerala

Skills

- **Data Visualization:** Microsoft Power BI, Excel, Matplotlib, Seaborn
- **Languages:** Python, C Programming, R Programming, SQL, HTML
- **Cloud Platforms:** Amazon Web Service, Google Cloud Platform
- **Software:** ArcGIS Pro, QGIS, Adobe PhotoShop, LaTeX, Canva, MS Office Suit, OpenStreetMap,
- **Data Analysis:** Numpy, Pandas, Rasterio, GDAL
- **Operating systems:** Windows, Ubuntu
- **Soft Skills:** Presentation, Planning, Organized, Creative Problem-Solving, Teamwork, Active Listening, Adaptability, Analytical Thinking